

B1 Assignment : Trigonometry

SINGLE CORRECT TYPE QUESTIONS

T01/1/001. If $\frac{(\cos 1^\circ + \sin 1^\circ)(\cos 2^\circ + \sin 2^\circ)(\cos 3^\circ + \sin 3^\circ) \dots (\cos 45^\circ + \sin 45^\circ)}{\cos 1^\circ \cdot \cos 2^\circ \cos 3^\circ \dots \cos 45^\circ} = a^b$, where a & b are prime

numbers and $a < b$, then $a + b$ is equal to :

- (A) 22 (B) 23 (C) 24 (D) 25

Ans. D

Sol.
$$\frac{(\cos 1^\circ + \cos 89^\circ)(\cos 2^\circ + \cos 88^\circ)(\cos 3^\circ + \cos 87^\circ) \dots (\cos 45^\circ + \cos 45^\circ)}{\cos 1^\circ \cdot \cos 2^\circ \cos 3^\circ \dots \cos 45^\circ}$$

$$= \frac{(2 \cos 45^\circ \cos 44^\circ)(2 \cos 45^\circ \cos 43^\circ)(2 \cos 45^\circ \cos 42^\circ) \dots 2 \cos 45^\circ}{\cos 1^\circ \cdot \cos 2^\circ \cos 3^\circ \dots \cos 45^\circ}$$

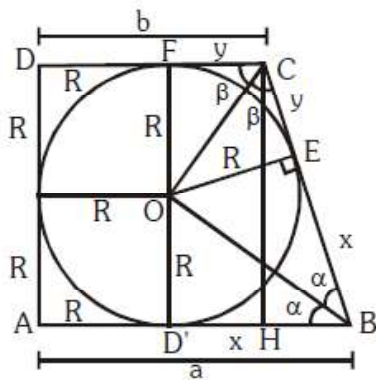
$$= (\sqrt{2})^{44} \cdot 2 = 2^{23} \Rightarrow a + b = 25$$

T01/1/002. A right angled trapezium is circumscribed about a circle. The radius of the circle, if the lengths of the bases (i.e. parallel sides) are equal to a and b is :

- (A) $a + b$ (B) $\frac{ab}{a + b}$ (C) $\frac{a + b}{ab}$ (D) $|a - 1 - b|$

Ans. B

Sol. Let ABCD is a right angled trapezium.
 Let radius of circle is R



& let $BD = BE = x$ and
 $CE = CF = y$
 $\therefore AB = a$

$$R + x = a \Rightarrow x = a - R \quad \dots(1)$$

$$\& DC = b$$

$$R + y = b \Rightarrow y = b - R \quad \dots(2)$$

$$\text{In } \triangle OD'B, \tan \alpha = \frac{R}{x}$$

$$\text{Now in } \triangle CHB, \sin 2\alpha = \frac{CH}{BC}$$

$$\Rightarrow \frac{2 \tan \alpha}{1 + \tan^2 \alpha} = \frac{2R}{x + y}$$

$$\Rightarrow \frac{\frac{2R}{x}}{1 + \frac{R^2}{x^2}} = \frac{2R}{x + y}$$

$$\Rightarrow \frac{2Rx}{x^2 + R^2} = \frac{2R}{x + y}$$

$$\Rightarrow x^2 + xy = x^2 + R^2 \Rightarrow xy = R^2 \Rightarrow (a - R)(b - R) = R^2 \quad [\text{From (1) and (2)}]$$

$$\Rightarrow ab - (a + b)R + R^2 = R^2$$

$$\Rightarrow ab - (a + b)R = 0$$

$$\therefore R = \frac{ab}{a + b}$$

T01/1/003. If A, B, C and D denotes the interior angles of a quadrilateral then :

$$(A) \frac{\prod \tan A}{\sum \cot A} = \sum \tan A$$

$$(B) \sum \tan A = (\prod \tan A)(\sum \cot A)$$

$$(C) \sum \cot A - \prod \tan A = \sum \tan A$$

$$(D) \sum \tan A = \prod \tan A - \sum \cot A$$

Ans. B

$$\text{Sol. } \tan(A + B + C + D) = \frac{S_1 - S_3}{1 - S_2 + S_4} \quad (\because A + B + C + D = 360^\circ)$$

$$S_1 = S_3$$

$$\tan A + \tan B + \tan C + \tan D = \prod \tan A (\sum \cot A)$$

T01/1/004. The expression $S = \sec 11^\circ \sec 19^\circ - 2 \cot 71^\circ$ reduces to :

- (A) $2 \cot 11^\circ$ (B) $\tan 19^\circ$ (C) $2 \tan 11^\circ$ (D) $\frac{1}{2} \tan 19^\circ$

Ans. C

Sol.

$$S = \frac{1}{\cos 11^\circ \cos 19^\circ} - \frac{2 \sin 19^\circ}{\cos 19^\circ}$$

$$= \frac{1 - 2 \sin 19^\circ \cos 11^\circ}{\cos 11^\circ \cos 19^\circ} = \frac{1 - (\sin 30^\circ + \sin 8^\circ)}{\cos 11^\circ \cos 19^\circ} = \frac{\left(\frac{1}{2}\right) - \sin 8^\circ}{\cos 11^\circ \cos 19^\circ} = \frac{\sin 30^\circ - \sin 8^\circ}{\cos 11^\circ \cos 19^\circ}$$

$$= \frac{2 \cos 19^\circ \sin 11^\circ}{\cos 11^\circ \cos 19^\circ} = 2 \tan 11^\circ$$

T01/1/005. The average of the numbers $n \sin n^\circ$ for $n = 2, 4, 6, \dots, 180$:

- (A) 1 (B) $\cot 1^\circ$ (C) $\tan 1^\circ$ (D) $\frac{1}{2}$

Ans. B

Sol. $S = 2 \sin 2^\circ + 4 \sin 4^\circ + \dots + 178 \sin 178^\circ + 180 \sin 180^\circ$

$$S = 2[\sin 2^\circ + 2 \sin 4^\circ + 3 \sin 6^\circ + \dots + 89 \sin 178^\circ]$$

$$\frac{S}{2S} = \frac{2[89 \sin 178^\circ + 88 \sin 176^\circ + \dots + 1 \sin 2^\circ]}{2[90(\sin 2^\circ + \sin 4^\circ + \sin 6^\circ + \dots + \sin 178^\circ)]}$$

$$S = 90 \cdot \frac{\sin\left(\frac{n\theta}{2}\right)}{\sin \frac{\theta}{2}} \sin\left(\frac{(n+1)\theta}{2}\right) = \frac{90 \sin(89^\circ)}{\sin 1^\circ} \cdot \sin 90^\circ \quad (\theta = 2^\circ)$$

$$S = 90^\circ \cot 1^\circ$$

$$\text{Average value} = \frac{90 \cot 1^\circ}{90} = \cot 1^\circ$$

T01/1/006. If a and b are real number such that $a^2 + ab < 0$, then the expression

$$\frac{1}{2\sqrt{-a^2-ab}} \left(\frac{\frac{\sqrt{-a^2-ab}}{\cos^2 x}}{a + \left(\sqrt{-a^2-ab}\right)\tan x} - \frac{\frac{\sqrt{-a^2-ab}}{\cos^2 x}}{-a + \left(\sqrt{-a^2-ab}\right)\tan x} \right) \text{ simplified to :}$$

$$(A) \frac{1}{a + b \cos^2 x}$$

$$(B) \frac{1}{a + b \sin^2 x}$$

$$(C) \frac{1}{a + b \sec^2 x}$$

(D) some constant independent of x

Ans. B

Sol. Let $\sqrt{-(a^2 + ab)} = p > 0 \Rightarrow p^2 = -(a^2 + ab)$

$$\begin{aligned} E &= \frac{1}{2p} \left(\frac{p \sec^2 x}{a + p \tan x} + \frac{p \sec^2 x}{a - p \tan x} \right) = \frac{p \sec^2 x}{2p} \left(\frac{(a - p \tan x) + (a + p \tan x)}{a^2 - p^2 \tan^2 x} \right) \\ &= \frac{\sec^2 x}{2} \left(\frac{2a}{a^2 + (a^2 + ab) \tan^2 x} \right) = \left(\frac{\sec^2 x}{a \sec^2 x + b \tan^2 x} \right) = \frac{1}{a + b \sin^2 x} \end{aligned}$$

T01/1/007. The expression $\sum_{k=0}^{10} 2^k \tan(2^k)$ equals :

$$(A) \cot 1 + 2^{11} \cot(2^{11})$$

$$(B) \cot 1 - 2^{10} \cot(2^{10})$$

$$(C) \cot 1 + 2^{10} \cot(2^{10})$$

$$(D) \cot 1 - 2^{11} \cot(2^{11})$$

Ans. D

Sol. We use the relation $\cos(x) = \frac{\sin(2x)}{2 \sin(x)}$

$$\prod_{k=0}^{10} \cos(2^k x) = \prod_{k=0}^{10} \frac{\sin(2^{k+1} x)}{2 \sin(2^k x)} = \frac{\prod_{k=1}^{11} \sin(2^k x)}{2^{11} \prod_{k=0}^{10} \sin(2^k x)} = \frac{\sin(2^{11} x)}{2^{11} \sin(x)}$$

Take the log and differentiate both sides of this equation :

$$-\sum_{k=0}^{10} 2^k \tan(2^k x) = 2^{11} \cot(2^{11} x) - \cot(x)$$

Hence,

$$\sum_{k=0}^{10} 2^k \tan(2^k) = \cot(1) - 2^{11} \cot(2^{11})$$

T01/1/008. The number of real roots of the equation $2 \cos\left(\frac{x^2 + x}{6}\right) = 2^x + 2^{-x}$ is :

- (A) 0 (B) 1 (C) 2 (D) Infinitely many

Ans. B

Sol. Minimum value of $2^x + 2^{-x} = 2$

$$\Rightarrow 2 \cos\left(\frac{x^2 + x}{6}\right) = 2^x + 2^{-x}$$

$$\Rightarrow 2 \cos\left(\frac{x^2 + x}{6}\right) \geq 2$$

Now, $\because \cos 0 = 1$

$$\Rightarrow \frac{x^2 + x}{6} = 0 \Rightarrow x(x + 1) = 0 \Rightarrow x = 0, x = -1$$

So, $x = 0$ is the only solution as the equation is not satisfying at $x = -1$

T01/1/009. Let $[x]$ denotes the greatest integer $\leq x$. If :

$$2 \sin(x + y) = \frac{2}{3} = \cos(x - y); -\frac{\pi}{4} < y < 0 < x < \frac{\pi}{4},$$

$$\lambda = \frac{\sin x}{\cos y} + \frac{\sin y}{\cos x} + \frac{\cos x}{\sin y} + \frac{\cos y}{\sin x},$$

Then $[\lambda^2]$ is equal to :

- (A) 1 (B) 2 (C) 3 (D) 4

Ans. A

Sol.
$$\lambda = \left(\frac{\sin x}{\cos y} + \frac{\cos x}{\sin y} \right) + \left(\frac{\sin y}{\cos x} + \frac{\cos y}{\sin x} \right)$$

$$= \frac{2 \cos(x - y)}{2 \sin y \cos y} + \frac{2 \cos(x - y)}{2 \sin x \cos x}$$

$$= 2 \cos(x-y) \left[\frac{1}{\sin 2y} + \frac{1}{\sin 2x} \right] \quad \dots(i)$$

Now,

$$\sin 2x = \sin((x+y) + (x-y)) = \sin(x+y) \cos(x-y) + \cos(x+y) \sin(x-y)$$

$$= \frac{1}{3} \cdot \frac{2}{3} + \frac{2\sqrt{2}}{3} \cdot \frac{\sqrt{5}}{3}$$

$$= \frac{2}{9}(\sqrt{10} + 1)$$

$$\sin 2y = \sin((x+y) - (x-y)) = \frac{2}{9}(1 - \sqrt{10})$$

Put in (i)

$$\lambda = \frac{4}{3} \left[\frac{9}{2(1-\sqrt{10})} + \frac{9}{2(1+\sqrt{10})} \right]$$

$$\lambda = \frac{2}{3} \left[\frac{9}{10-1}(\sqrt{10}-1) - \frac{9}{10-1}(\sqrt{10}+1) \right]$$

$$\lambda = \frac{2}{3}[-2]$$

$$\lambda = -\frac{4}{3} \Rightarrow \lambda^2 = \frac{16}{9}$$

$$\Rightarrow [\lambda^2] = 1$$

T02/1/010. Number of solutions of $\sec x + \cot x = \cos x + \tan x$ in $[0, 2\pi]$ is equal to :

- (A) 0 (B) 1 (C) 2 (D) 3

Ans. C

Sol. $\sec x - \cos x = \tan x - \cot x$

$$\frac{\sin^2 x}{\cos x} = \frac{2 \sin^2 x - 1}{\sin x \cos x}$$

$$\Rightarrow \sin^3 x = 2 \sin^2 x - 1 \quad \left(\because \cos x \neq 0 \Rightarrow x \neq \frac{\pi}{2}, \frac{3\pi}{2} \right)$$

$$\Rightarrow (\sin x - 1)(\sin^2 x - \sin x - 1) = 0$$

$$\Rightarrow \sin x = 1 \quad \Rightarrow \text{rejected} \quad \left(\because x \neq \frac{\pi}{2}, \frac{3\pi}{2} \right)$$

$$\text{Now } \sin^2 x - \sin x - 1 = 0 \quad \Rightarrow \sin x = \frac{1 \pm \sqrt{5}}{2}, \sin x = \frac{1 + \sqrt{5}}{2} \rightarrow \text{Rejected} \left(\because \frac{1 + \sqrt{5}}{2} > 1 \right)$$

$\Rightarrow$ Two solutions.

T02/1/011. Least positive solution of equation $(\sqrt{3} \sin x - \cos x)(\sqrt{3} \cos x + \sin x) = 2 \sin 10x$ is equal to :

- (A) $\frac{\pi}{9}$ (B) $\frac{5\pi}{24}$ (C) $\frac{\pi}{18}$ (D) $\frac{7\pi}{24}$

Ans. A

Sol.
$$\left(\frac{\sqrt{3}}{2} \sin x - \frac{1}{2} \cos x \right) \left(\frac{\sqrt{3}}{2} \cos x + \frac{1}{2} \sin x \right) = \frac{\sin 10x}{2}$$

$$\sin \left(x - \frac{\pi}{6} \right) \cos \left(x - \frac{\pi}{6} \right) = \frac{\sin 10x}{2}$$

$$\Rightarrow \sin \left(2x - \frac{\pi}{3} \right) = \sin 10x$$

$$\Rightarrow 2 \cos \left(6x - \frac{\pi}{6} \right) \sin \left(4x + \frac{\pi}{6} \right) = 0$$

$$\Rightarrow 6x - \frac{\pi}{6} = (2n+1)\frac{\pi}{2} \text{ or } 4x + \frac{\pi}{6} = k\pi$$

for $n = 0$ and $k = 1$

$$6x = \frac{\pi}{2} + \frac{\pi}{6} \text{ or } 4x = \pi - \frac{\pi}{6}$$

$$x = \frac{\pi}{9} \text{ or } \frac{5\pi}{24}$$

least positive solution is $\frac{\pi}{9}$

T02/1/012. Number of ordered pairs (x, y) satisfying the conditions $\sin^{22}x + \cos^{22}x = \sec^2y - 2\sec y + 2$ and $x, y \in [0, 2\pi]$ is equal to :

- (A) 8 (B) 10 (C) 6 (D) 4

Ans. A

Sol. $\sin^{22}x + \cos^{22}x = \sec^2y - 2\sec y + 2$

$$\text{L.H.S.} \leq 1 \quad \text{R.H.S.} = (\sec y - 1)^2 + 1 \geq 1$$

If L.H.S. = R.H.S.

$$\Rightarrow \sin^2 x + \cos^2 x = 1 \text{ and } \sec y = 1$$

$$\downarrow \quad \downarrow$$

$$x = \frac{\pi}{2}, \frac{3\pi}{2} \quad x = 0, 2\pi \quad y = 0, 2\pi$$

T02/1/013. The general solution of x for which $\sqrt{3} \sin x - \cos x = \min_{\lambda \in \mathbb{R}} \{2, e^2, \pi, \lambda^2 - 4\lambda + 7\}$ are given by :

(A) $2n\pi, n \in \mathbb{I}$

(B) $2n\pi + \frac{2\pi}{3}, n \in \mathbb{I}$

(C) $n\pi + (-1)^n \frac{\pi}{4} + \frac{\pi}{6}, n \in \mathbb{I}$

(D) $n\pi + (-1)^{n+1} \frac{\pi}{4} - \frac{\pi}{3}, n \in \mathbb{I}$

Ans. B

Sol. $\sqrt{3} \sin x - \cos x = \min \{2, e^2, \pi, (\lambda - 2)^2 + 3\} = 2$

$$\{2, e^2, \pi, \lambda^2 - 4\lambda + 7\} = \{2, > 2, > 2, (\lambda - 2)^2 + 3\}$$

$$\{2, > 2, > 2, > 2\}$$

$$\Rightarrow \sqrt{3} \sin x - \cos x = 2$$

$$\Rightarrow \frac{\sqrt{3}}{2} \sin x - \frac{1}{2} \cos x = 1$$

$$\Rightarrow \sin x \cdot \sin \frac{\pi}{3} - \cos x \cos \frac{\pi}{3} = 1$$

$$\Rightarrow \cos \left(x + \frac{\pi}{3} \right) = -1$$

$$\Rightarrow x + \frac{\pi}{3} = (2n+1)\pi$$

$$x = (2n+1)\pi - \frac{\pi}{3}$$

T02/1/014. Number of the points of intersection of the curves $y = 2\sin 2x$ and $y = 3(\sin x + \cos x - 1)$, if $0 \leq x \leq 2\pi$:

(A) 4

(B) 5

(C) 7

(D) 8

Ans. B

Sol. $2 \sin 2x = 3(\sin x + \cos x - 1)$

$$2((\sin x + \cos x)^2 - 1) = 3(\sin x + \cos x) - 3$$

$$\text{Let } \sin x + \cos x = t$$

$$2t^2 - 3t + 1 = 0$$

$$\Rightarrow t = \frac{1}{2} \text{ or } t = 1$$

$$\Rightarrow \sin x + \cos x = \frac{1}{2} \text{ or } \sin x + \cos x = 1$$

$\Downarrow$

$\Downarrow$

$$\sin\left(x + \frac{\pi}{4}\right) = \frac{1}{2\sqrt{2}} \quad \text{three solutions } 0, \frac{\pi}{2}, 2\pi$$

$\Downarrow$

Two solutions

$\therefore$ total 5 solutions

T02/1/015. Sum of all the solutions of the inequation $\sqrt{2 - \cos^7 x} \leq \sin^{19} x + \cos^{21} x$ in the interval $[-2\pi, 2\pi]$ is given by:

- (A) 0 (B) 3 (C) 2π (D) $\frac{19\pi}{2}$

Ans. A

Sol. We know that

$$\sin^2 x \geq \sin^{19} x$$

$$\cos^2 x \geq \cos^{21} x$$

$$\Rightarrow \sin^2 x + \cos^2 x \geq \sin^{19} x + \cos^{21} x$$

$$\Rightarrow \text{R.H.S } \sin^{19} x + \cos^{21} x \leq 1 \text{ but L.H.S.}$$

$$\sqrt{2 - \cos^7 x} \geq 1$$

$$\text{so L.H.S} = \text{R.H.S} = 1$$

$$\Rightarrow \sin^{19} x + \cos^{21} x = 1 \Rightarrow \text{if } \sin x = 0$$

$$\Rightarrow \cos x = 1 \Rightarrow x = 0, 2\pi, -2\pi$$

$$\text{and if } \sin x = 1 \Rightarrow \cos x = 0 \Rightarrow -\frac{3\pi}{2}, +\frac{\pi}{2} \quad \dots(1)$$

$$\text{and } 2 - \cos^7 x = 1 \Rightarrow \cos^7 x = 1$$

$$\Rightarrow x = -2\pi, 0, 2\pi \quad \dots(2)$$

from (1) and (2) solutions are $0, 2\pi, -2\pi$

ONE OR MORE THAN ONE CORRECT TYPE QUESTIONS

T01/2/016. The value of $\text{antilog}_5 \left[\frac{\tan^2\left(\frac{\pi}{5}\right) + \tan^2\left(\frac{2\pi}{5}\right) + 20}{\cot^2\left(\frac{\pi}{5}\right) + \cot^2\left(\frac{2\pi}{5}\right) + 28} \right]$ is equal to :

- (A) odd number (B) even number (C) prime number (D) composite number

Ans. AC

Sol. $\cot^2 36^\circ + \tan^2 18^\circ$

$$\frac{\left(\frac{\sqrt{5}+1}{4}\right)^2}{1-\left(\frac{\sqrt{5}+1}{4}\right)^2} + \frac{\left(\frac{\sqrt{5}-1}{4}\right)^2}{1-\left(\frac{\sqrt{5}-1}{4}\right)^2} = 2$$

T01/2/017. If $\sin^4 \theta + \frac{1}{\sin^4 \theta} = 194$, then the value of $(2\text{cosec}\theta - \cot\theta \cos\theta)$ can be :

- (A) 2 (B) -2 (C) 4 (D) -4

Ans. CD

Sol. $\sin^4 \theta + \frac{1}{\sin^4 \theta} = 194$

$$\left(\sin^2 \theta + \frac{1}{\sin^2 \theta}\right)^2 = 194 + 2 = 196$$

$$\sin^2 \theta + \frac{1}{\sin^2 \theta} = 14$$

$$\left(\sin \theta + \frac{1}{\sin \theta}\right)^2 = 14 + 2 = 16$$

$$\sin \theta + \frac{1}{\sin \theta} = 4 \quad \text{or} \quad \sin \theta + \frac{1}{\sin \theta} = -4$$

$$2 \text{ cosec}\theta - \cot\theta \cos\theta = \frac{2 - \cos^2 \theta}{\sin \theta}$$

$$= \frac{1 + \sin^2 \theta}{\sin \theta} = \sin \theta + \frac{1}{\sin \theta}$$

T01/2/018. If equations $bx^2 + acx + b^2c = 0$ and $cx^2 + abx + b^2c = 0$ have a common root, (where a, b, c are non-zero distinct real numbers), then which of the following is/are correct :

(A) $\frac{a}{c} + \frac{b}{a} + \frac{a}{b} = 0$

(B) $\frac{a}{b} + \frac{b}{c} + \frac{c}{a} = 0$

(C) $\frac{1}{bc^2} + \frac{1}{a^2c} + \frac{1}{cb^2} = 0$

(D) $\frac{1}{ac^2} + \frac{1}{ba^2} + \frac{1}{cb^2} = 0$

Ans. AC

Sol. Solving we get common root, $\alpha = a$

condition of common root,

$$a^2b + a^2c + b^2c = 0 \quad \text{.....(i)}$$

on dividing equation (i) by abc, we get

$$\frac{a}{c} + \frac{a}{b} + \frac{b}{a} = 0$$

on dividing equation (i) by $a^2b^2c^2$, we get

$$\frac{1}{bc^2} + \frac{1}{cb^2} + \frac{1}{a^2c} = 0$$

T01/2/019. Let $S(n) = \sum_{r=1}^n \frac{\cos(2 \cdot 3^{r-1} \alpha)}{\sin(3^r \alpha)}$, $\alpha = 18^\circ$ and $P(n) = \sum_{r=1}^n \frac{\cos(3 \cdot 2^{r-1} \beta)}{\cos(2^r \beta)}$, $\beta = 36^\circ$, then :

(A) $S(2012) + P(2012) = 2010$

(B) $S(2012) + P(2012) = 2012$

(C) $P(2011) - S(2011) = 2009$

(D) $P(2011) - S(2011) = 2008$

Ans. AD

Sol. $2\alpha + 3\alpha = 90^\circ$

$$2\alpha = 90^\circ - 3\alpha$$

$$2 \cdot 3^{r-1} \alpha = (90^\circ) \times (3^{r-1}) - 3^r \alpha$$

$$\cos(2 \cdot 3^{r-1} \alpha) = \pm \sin(3^r \alpha)$$

$$\Rightarrow \frac{\cos(2 \cdot 3^{r-1} \alpha)}{\sin(3^r \alpha)} = \begin{cases} 1, & r \text{ is odd} \\ -1, & r \text{ is even} \end{cases}$$

$$\text{Similarly, } \cos(2^r \beta) = \begin{cases} -\cos(3 \cdot 2^{r-1} \beta), & r = 1 \\ \cos(3 \cdot 2^{r-1} \beta), & r \geq 2 \end{cases}$$

T01/2/020. Let $E = \frac{2 \cos 1^\circ \cos 2^\circ \sin 86^\circ}{\cos \frac{98^\circ}{3} \cos \frac{82^\circ}{3} \cos \frac{262^\circ}{3}}$, then :

- (A) $E > 2$ (B) $E < 2$ (C) $E = \operatorname{cosec} 1^\circ$ (D) $E = \sec 1^\circ$

Ans. AC

Sol.
$$E = \frac{2 \cos 1^\circ \cos 2^\circ \cos 4^\circ}{\cos \left(60^\circ - \frac{82^\circ}{3} \right) \cos \frac{82^\circ}{3} \cos \left(60^\circ + \frac{82^\circ}{3} \right)}$$

$$= \frac{2 \times \sin 8^\circ}{2^3 \sin 1^\circ} \times \frac{1}{\frac{1}{4} \times \cos 82^\circ} = \operatorname{cosec} 1^\circ$$

$$\operatorname{cosec} 1^\circ > \operatorname{cosec} \frac{\pi}{6} \Rightarrow \operatorname{cosec} 1^\circ > 2$$

$\Rightarrow$ Option (A) & (C) are correct.

T01/2/021. Let $\theta \in \left(0, \frac{\pi}{4} \right)$ and $I_1 = \sin((\tan \theta)^{\cot \theta})$, $I_2 = \sin((\sin \theta)^{\cot \theta})$, $I_3 = \tan((\tan \theta)^{\cot \theta})$ and $I_4 = \tan((\sin \theta)^{\cot \theta})$,

then which is/are always true :

- (A) $I_1 > I_2$ (B) $I_4 > I_3$ (C) $I_3 > I_1$ (D) $I_3 > I_2$

Ans. ACD

Sol. in $\theta \in \left(0, \frac{\pi}{6} \right)$ $0 < \sin \theta < \tan \theta < 1 < \cot \theta$

$$\sin \theta < \tan \theta \Rightarrow (\sin \theta)^{\cot \theta} < (\tan \theta)^{\cot \theta}$$

$$\Rightarrow \sin((\sin \theta)^{\cot \theta}) < \sin((\tan \theta)^{\cot \theta}) \Rightarrow I_1 > I_2 \quad \dots(i)$$

$$\because (\tan \theta)^{\cot \theta} < 1$$

$$\Rightarrow \sin((\tan \theta)^{\cot \theta}) < \tan((\tan \theta)^{\cot \theta}) \Rightarrow I_3 > I_1 \quad \dots(ii)$$

from (i) and (ii) $I_3 > I_1 > I_2$

T01/2/022. Let $\alpha, \beta + \sin \theta$ are roots of $x^2 + ax + a - 1 = 0$, $\alpha - \cos \theta$ and β are roots of $x^2 - 3x - a^2 + 3a = 0$, $\beta > \alpha$

where $\theta \in \left(0, \frac{\pi}{2} \right)$, if N is number of possible integral value(s) of a and P is minimum integral value of

$\alpha \sin \theta + \beta \cos \theta$, then which of the following option(s) is (are) true ?

(A) $N = 0$

(B) $N = 1$

(C) $|P|$ is prime number

(D) $|P|$ is composite number

Ans.

AD

Sol.

$$\alpha + \beta + \sin \theta = -a \quad \dots(i)$$

$$\alpha\beta + \alpha \sin \theta = a - 1 \quad \dots(ii)$$

$$\alpha + \beta - \cos \theta = 3 \quad \dots(iii)$$

$$\alpha\beta - \beta \cos \theta = 3a - a^2 \quad \dots(iv)$$

$$(i)-(iii) \quad \sin \theta + \cos \theta = -a - 3$$

$$\Rightarrow a = -3 - (\cos \theta + \sin \theta)$$

$$a \in [-3 - \sqrt{2}, -4) \quad \because \theta \in \left(0, \frac{\pi}{2}\right) \Rightarrow N = 0$$

$$(ii)-(iv) \quad \alpha \sin \theta + \beta \cos \theta = a^2 - 2a - 1$$

$$= (a - 1)^2 - 2$$

$$\therefore a \in [-3 - \sqrt{2}, -4)$$

$$\Rightarrow \text{range of } \alpha \sin \theta + \beta \cos \theta \text{ is } (23, 18 + 8\sqrt{2}]$$

$$\therefore P = 24$$

T01/2/023. Let θ, α satisfy a simultaneous set of equations given by $\tan \theta + \tan \alpha = 2$ & $\sin 2\theta = \cos \alpha$, then identify the correct statement(s) (where $\theta, \alpha \in [0, 2\pi]$):

(A) Number of ordered pairs (θ, α) is 4

(B) Number of ordered pairs (θ, α) is 8

(C) $\exists \theta, \alpha$ such that $\alpha = 4\theta$

(D) $\exists \theta, \alpha$ such that $\alpha = 2\theta$

Ans.

BC

Sol.

$$\sin 2\theta = \cos \alpha$$

$$\Rightarrow \cos \alpha = \cos \left(\frac{\pi}{2} - 2\theta \right)$$

$$\Rightarrow \alpha = 2n\pi \pm \left(\frac{\pi}{2} - 2\theta \right)$$

$$\text{Case -1 : } \alpha = 2n\pi + \frac{\pi}{2} - 2\theta$$

$$\Rightarrow \tan \alpha = \cot 2\theta$$

$$\text{Let } \tan \theta = t$$

$$\Rightarrow \tan \theta + \tan \theta = 2 \quad \Rightarrow t + \frac{1-t^2}{2t} = 2$$

$$\Rightarrow t^2 - 4t + 1 = 0 \quad \Rightarrow t = 2 \pm \sqrt{3}$$

$$\Rightarrow \theta, \alpha \in (15^\circ, 60^\circ), (75^\circ, 300^\circ), (195^\circ, 60^\circ), (255^\circ, 300^\circ)$$

Similarly for the second case, number of ordered pairs = 4

T01/2/024. If each pair of equations $x^2 + \sin A \cdot x - \sin^2 A = 0$, $x^2 + \sin B \cdot x - \sin^2 B = 0$ & $x^2 + \sin C \cdot x - \sin^2 C = 0$ quadratic in x has exactly one root common, then sum of roots of all three equations is equal to (Provided that $\sin A \neq \sin B$, $\sin B \neq \sin C$ & $\sin C \neq \sin A$):

(A) $-(\sin A + \sin B + \sin C)$

(B) $4(\sin A + \sin B + \sin C)$

(C) $2(\sin A + \sin B + \sin C)$

(D) 0

Ans. ABCD

Sol. Let roots of $x^2 + \sin A x - \sin^2 A = 0$ be α, β ... (1)

and of $x^2 + \sin B x - \sin^2 B = 0$ be β, γ ... (2)

and for $x^2 + \sin C x - \sin^2 C = 0$ be γ, α ... (3)

then using sum of roots

$$2(\alpha + \beta + \gamma) = -(\sin A + \sin B + \sin C) \quad \dots (4)$$

but from (1) - (2) $\Rightarrow \beta = \sin A + \sin B$

$$(2) - (3) \Rightarrow \gamma = \sin A + \sin C$$

$$(3) - (1) \Rightarrow \alpha = \sin C + \sin A$$

$$\Rightarrow \alpha + \beta + \gamma = 2(\sin A + \sin B + \sin C) \quad \dots (5)$$

from (4) and (5)

$$\alpha + \beta + \gamma = 0$$

T02/2/025. $\sqrt{\cos 2x} + \sqrt{1 + \sin 2x} = 2\sqrt{\sin x + \cos x}$, ($n \in \mathbb{I}$) if:

(A) $\sin x + \cos x = 0$

(B) $x = 2n\pi$

(C) $x = n\pi - \frac{\pi}{4}$

(D) None of these

Ans. ABC

Sol. $\sqrt{\cos 2x} + \sqrt{1 + \sin 2x} = 2\sqrt{\sin x + \cos x}$

$$\Rightarrow \sqrt{(\cos^2 x - \sin^2 x)} + \sqrt{(\cos x + \sin x)^2} = 2\sqrt{\cos x + \sin x}$$

$$\Rightarrow \cos x + \sin x = 0$$

$$\Rightarrow \tan x = -1 \Rightarrow x = n\pi - \frac{\pi}{4}$$

$$\text{Now, } \sqrt{\cos x + \sin x} \{ \sqrt{\cos x - \sin x} + \sqrt{\cos x + \sin x} - 2 \} = 0$$

$$\Rightarrow \cos x + \sqrt{\cos 2x} = 2$$

$$\Rightarrow \cos^2 x + 4 \cos x - 5 = 0$$

$$\Rightarrow \cos x = 1$$

$$\Rightarrow x = 2n\pi$$

T02/2/026. If $(\sin 2x + \sqrt{3} \cos 2x)^2 - 5 = \cos\left(2x - \frac{\pi}{6}\right)$ then the value of $\cos x$ can be :

(A) $\frac{\sqrt{6} + \sqrt{2}}{4}$ (B) $\frac{\sqrt{6} - \sqrt{2}}{4}$ (C) $-\left(\frac{\sqrt{6} + \sqrt{2}}{4}\right)$ (D) $\frac{\sqrt{2} - \sqrt{6}}{4}$

Ans. BCD

Sol. The given equation is $4\left(\frac{\sin 2x}{2} + \frac{\sqrt{3}}{2} \cos 2x\right)^2 = 5 + \cos\left(2x - \frac{\pi}{6}\right)$

$$\text{or } 4\left(\cos 2x \cos \frac{\pi}{6} + \sin 2x \cdot \sin \frac{\pi}{6}\right)^2 = 5 + \cos\left(2x - \frac{\pi}{6}\right) \text{ or } 4 \cos^2\left(2x - \frac{\pi}{6}\right) - \cos\left(2x - \frac{\pi}{6}\right) - 5 = 0$$

$$\text{Let } \cos\left(2x - \frac{\pi}{6}\right) = t \Rightarrow 4t^2 - t - 5 = 0 \Rightarrow (4t - 5)(t + 1) = 0$$

$$\therefore t \neq \frac{5}{4} \text{ hence } t = -1$$

$$\cos\left(2x - \frac{\pi}{6}\right) = \cos(2n - 1)\pi$$

$$2x - \frac{\pi}{6} = (2n - 1)\pi \Rightarrow x = (2n - 1)\frac{\pi}{2} + \frac{\pi}{12} \Rightarrow x = n\pi - \frac{5\pi}{12}, n \in I$$

Now verify all the alternatives.

T02/2/027. Consider the following matrix formed by functions f and g . $[p_{ij}]_{1 \leq i, j \leq 2} = \begin{bmatrix} (f(x))^{f(x)} & (f(x))^{g(x)} \\ (g(x))^{f(x)} & (g(x))^{g(x)} \end{bmatrix}$ Then which of the following options is/are correct :

(A) If $f(x) = \tan x$, $g(x) = \cot x$, then $p_{22} > p_{21} > p_{11} > p_{12} \forall x \in \left(0, \frac{\pi}{4}\right)$

(B) If $f(x) = \tan x$, $g(x) = \cot x$, then $P_{22} < P_{21} < P_{11} < P_{12} \forall x \in \left(\frac{\pi}{4}, \frac{\pi}{2}\right)$

(C) If $f(x) = \sin x$, $g(x) = \cos x$, then $P_{21} > P_{22} > P_{11} > P_{12} \forall x \in \left(0, \frac{\pi}{4}\right)$

(D) If $f(x) = \sin x$, $g(x) = \cos x$, then $P_{21} < P_{22} < P_{11} < P_{12} \forall x \in \left(\frac{\pi}{4}, \frac{\pi}{2}\right)$

Ans. ACD

Sol. (A), (B)

$$P_{11} = (\tan x)^{\tan x}$$

$$P_{12} = (\tan x)^{\cot x}$$

$$P_{21} = (\cot x)^{\tan x}$$

$$P_{22} = (\cot x)^{\cot x}$$

$$x \in \left(0, \frac{\pi}{4}\right), \tan x < 1 \text{ \& } \cot x > 1$$

$$P_{22} > P_{21} > P_{11} > P_{12}$$

$$\forall x \in \left(\frac{\pi}{4}, \frac{\pi}{2}\right), \tan x > 1 \text{ \& } \cot x < 1$$

$$P_{11} > P_{12} > P_{22} > P_{21}$$

INTEGER TYPE QUESTIONS

T02/4/028. The sum of the solutions of the equation $4\sin^2 x + 8\cos^2 x - 4\sin x - 8\sqrt{3} \cos x + 7 = 0$ in $[0, 3\pi]$ is $k\pi/3$, then k is _____.

Ans. 7

Sol. $4\sin^2 x - 4\sin x + 8(\cos^2 x - \sqrt{3} \cos x) + 7 = 0$

$$(2\sin x - 1)^2 + 8\left(\cos x - \frac{\sqrt{3}}{2}\right)^2 = 0$$

$$\Rightarrow \sin x = \frac{1}{2}$$

$$\cos x = \frac{\sqrt{3}}{2}$$

$$\Rightarrow x = \frac{\pi}{6} \text{ and } \frac{13\pi}{6}$$

$$\Rightarrow \text{sum} = \frac{7\pi}{3} = \frac{k\pi}{3}$$

$$\Rightarrow k = 7$$

T02/4/029. Number of integral values of x in $(-\pi, \pi)$ for which $2x^4 \leq \cos^6 x + \sin^4 x - 1$ is _____.

Ans. 1

Sol.

$$2x^4 \leq \cos^6 x + \sin^4 x - 1$$

$$2x^4 \leq \cos^6 x + (\sin^2 x + 1)(-\cos^2 x)$$

$$2x^4 \leq \cos^2 x (\cos^4 x - 1 - \sin^2 x)$$

$$2x^4 \leq \cos^2 x ((\cos^2 x + 1)(-\sin^2 x) - \sin^2 x)$$

$$\underbrace{2x^4}_{\substack{\text{always +ve} \\ \text{or zero}}} \leq \underbrace{-\cos^2 x \cdot \sin^2 x (\cos^2 x + 2)}_{\text{always -ve or zero}}$$

$\Rightarrow x = 0$ will be only possible value.

T02/4/030. If $|x| + y = 3$ and $\left| \sin \frac{\pi x^2}{2} \right| = 1$, then number of ordered pairs (x, y) is (where y is a non negative integer) _____.

Ans. 4

Sol.

$$|x| + y = 3 \text{ and } \left| \sin \frac{\pi x^2}{2} \right| = 1$$

$$\Rightarrow x = \pm 1 \text{ and } y = 2$$

$$\text{or } x = \pm 3 \text{ and } y = 0$$

$$\Rightarrow 4 \text{ ordered pairs}$$

T02/4/031. If sum of all the solutions in $[0, 2\pi]$ of the equation $\sin(\sqrt{1 + \sin 2\theta}) = (\sin \theta + \cos \theta)$ is ' $k\pi$ ' then $(2k)$ is _____.

Ans. 5



$$\sin \sqrt{1 + \sin 2\theta} = (\sin \theta + \cos \theta)$$

$$\sin(|\sin \theta + \cos \theta|) = \sin \theta + \cos \theta$$

$$\text{Let } \sin \theta + \cos \theta = x$$

$$\sin|x| = x \quad \Rightarrow x = 0$$

$$\sin \theta + \cos \theta = 0 \quad \sin\left(\theta + \frac{\pi}{4}\right) = 0$$

$$\theta + \frac{\pi}{4} = n\pi \Rightarrow \theta = n\pi - \frac{\pi}{4} \Rightarrow \theta = \pi - \frac{\pi}{4}, 2\pi - \frac{\pi}{4}$$

$$\text{Sum of solutions} = \frac{5\pi}{2}$$

$$\Rightarrow k = \left(\frac{5}{2}\right) \Rightarrow 2k = 5$$